NHANES DATA ANALYSIS

Introduction

This project analyzes the health and nutritional status of adults in the United States using a combined dataset from three National Health and Nutrition Examination Survey (NHANES) files—demographic, examination, and laboratory—comprising 5,367 individuals aged 18 to 80. NHANES is conducted by the CDC's National Center for Health Statistics and provides comprehensive measures of health and nutrition across the U.S. population.

The dataset includes 16 variables: 10 health indicators—body measures (height, weight, waist circumference, BMI), cardiovascular indicators (systolic and diastolic blood pressure), and metabolic biomarkers (cholesterol, creatinine, glycohemoglobin); 5 demographic factors (age, gender, race/ethnicity, education, income); and 1 respondent identifier (SEQN). The central research questions of this report are **what latent structures or patterns exist among the health indicators,** and second, **how do health indicators relate to demographic features.**

Data Inspection

Missing values: Several laboratory variables exhibit moderate missingness (approximately 3.5–5%), and demographic variables such as education and income show non-response rates of 5.5–7.5%. Because many methods are sensitive to the structure of the covariance matrix, this analysis uses complete cases (4,931 individuals). However, this choice reduces generalizability: individuals with missing information may differ systematically from those retained.

Outliers: Outlier detection was performed using the Mahalanobis distance with a chi-square threshold of 0.975 ($p = 16$), identifying 345 multivariate outliers. Variable-specific outliers were also assessed via z-scores (threshold = 4), with glycohemoglobin showing the highest number of outliers (79). The prevalence of outliers is consistent with the heavy-tailed, non-normal distribution of certain health indicators, reflecting the heterogeneity and extreme cases observed in real-world populations—for example, individuals with particularly high blood sugar. Rather than removing these observations, the analysis retains all cases to preserve the integrity of the population structure and to ensure that downstream analyses capture these meaningful variations. The presence of numerous outliers also motivates the potential use of robust techniques, such as Minimum Covariance Determinant (MCD), in later analyses, as extreme values can disproportionately affect estimates of mean and covariance used in methods like Mahalanobis distance.

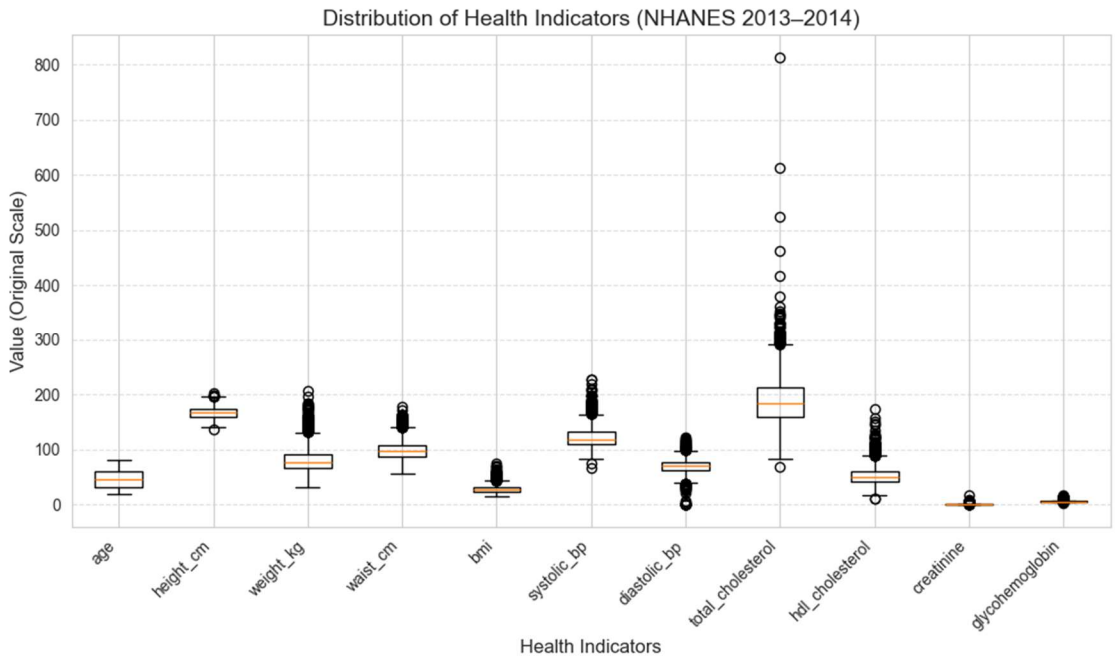


Figure 1. Boxplot of 11 numerical health indicators from the NHANES 2013–2014 dataset (n = 4931). The plot shows distributions, highlighting outliers and skewness. Weight, waist, BMI, systolic blood pressure, total cholesterol, and HDL cholesterol exhibit right-skewed distributions with numerous high-value outliers, reflecting the presence of individuals with obesity or elevated cholesterol. Creatinine and glycohemoglobin show smaller variance, indicating more homogeneous distributions. Differences in variability and skewness suggest latent structures, motivating the use of multivariate methods such as PCA, Factor Analysis, or Independent Component Analysis. Standardization was applied to control for scale differences, and deviations from multivariate normality indicate sensitivity to extreme values, justifying the consideration of robust or non-Gaussian methods.

Boxplot:

The boxplot (Figure 1) illustrates the distribution of the 11 numerical health indicators. Variables such as weight, waist, BMI, systolic blood pressure, total cholesterol, and HDL cholesterol exhibit numerous outliers above the upper quartile, indicating right-skewed distributions. This pattern likely reflects the presence of individuals with obesity or elevated cholesterol levels, highlighting real-world heterogeneity in the population. In particular, total cholesterol shows extreme high values, emphasizing the existence of high-risk individuals among the 4931 surveyed. Conversely, variables such as creatinine and glycohemoglobin display smaller variance, suggesting more homogeneous distributions.

The variability differences and skewed distributions across features imply that the dataset may be driven by latent structures, which motivates the application of Factor Analysis (FA) or Independent Component Analysis (ICA) to uncover underlying dimensions. Because the features are measured on different scales, standardization was applied prior to multivariate analysis to prevent features with larger magnitudes from disproportionately influencing results.

The presence of outliers and heteroskedasticity also indicates deviations from multivariate normality, implying that methods relying on covariance structures, such as PCA and FA, may be sensitive to noise and extreme values. This observation further motivates the careful assessment of distributional

assumptions and justifies considering alternative approaches, such as Kernel PCA or ICA, which are more robust to non-Gaussianity and potential nonlinear relationships.

Correlation Matrix, Assumption Testing and Method Selection:

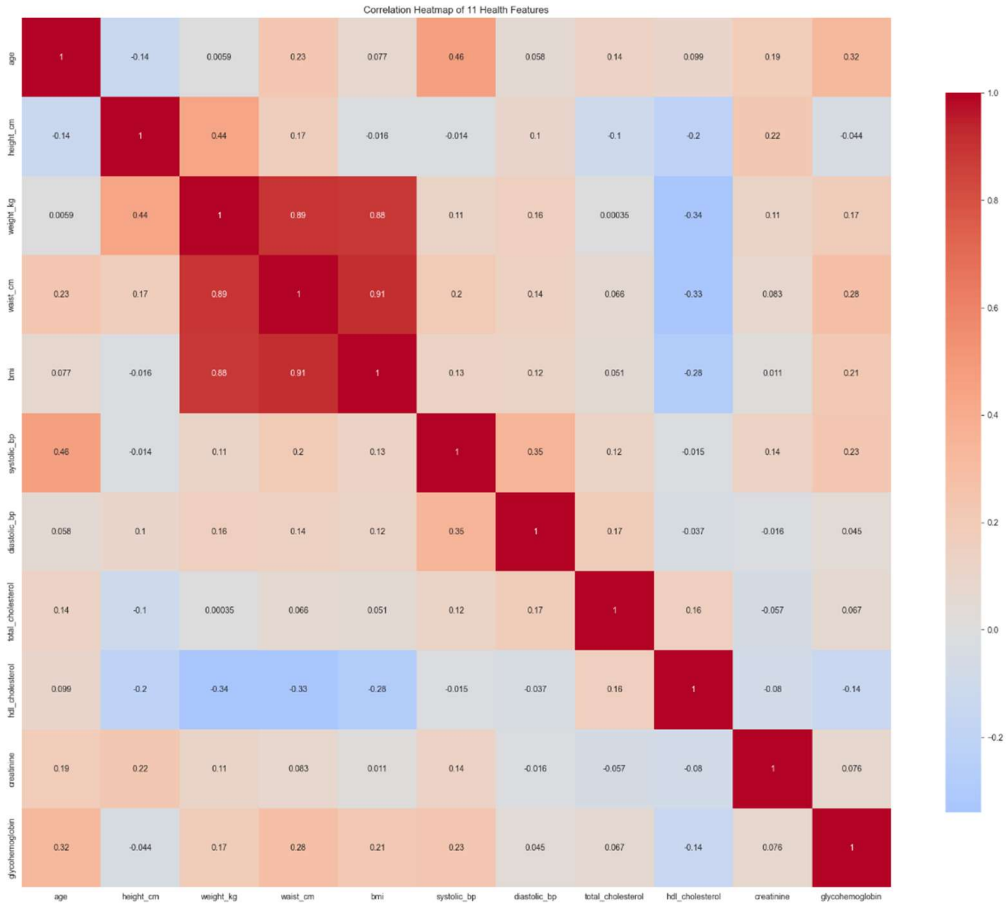


Figure 2. Correlation heatmap of the 11 health indicators.
The heatmap reveals several clear correlation patterns. Weight, waist circumference, and BMI form a strong positively correlated block (>0.85), indicating a shared body-composition dimension. Systolic and diastolic blood pressure show only moderate correlation (~0.35), suggesting partially distinct physiological processes. HDL cholesterol is negatively correlated with the body-measurement block, consistent with higher adiposity being linked to lower HDL. In contrast, creatinine and glycohemoglobin show weak or near-zero correlations with most variables, indicating more independent sources of variation. Overall, the structure combines strongly correlated groups (appropriate for PCA/FA) with largely independent variables (motivating ICA), supporting the use of multiple complementary multivariate methods.

Correlation matrix

The correlation matrix (Figure 2) reveals a clear block structure among a subset of variables. In particular, weight, waist circumference, and BMI exhibit strong correlations (all above 0.85). This is possible, as these variables are all body measurements. The strength of these relationships suggests the presence of a dominant latent factor related to overall body mass. This provides motivation for applying dimension reduction techniques such as Principal Component Analysis (PCA) to capture this shared variation efficiently, as well as Factor Analysis (FA) to interpret it as a latent construct.

In contrast, systolic and diastolic blood pressure show only moderate correlation (≈0.35), suggesting that they may load on different latent factors and capture different health measurements. Additionally, HDL cholesterol is negatively correlated with the body measurement block, reflecting known epidemiological patterns in which obesity is associated with lower "good" cholesterol levels. These relationships further support the use of FA, which allows overlapping contributions of variables to multiple latent factors.

In addition, several variables—particularly creatinine and glycohemoglobin—exhibit weak or near-zero correlations with most other features. This lack of linear dependence suggests that not all variation in the dataset can be explained by a small number of correlated latent factors. There may exists more independent sources. This observation motivates the use of Independent Component Analysis (ICA), which is specifically designed to identify statistically independent components and can capture structure that PCA and FA may overlook.

Overall, the correlation structure suggests a heterogeneous data-generating process: part of the variation arises from strongly correlated groups of variables (well-suited for PCA/FA), while other parts appear more independent (better captured by ICA). This justifies the use of multiple complementary methods to uncover different aspects of the underlying structure.

Assumption Testing:

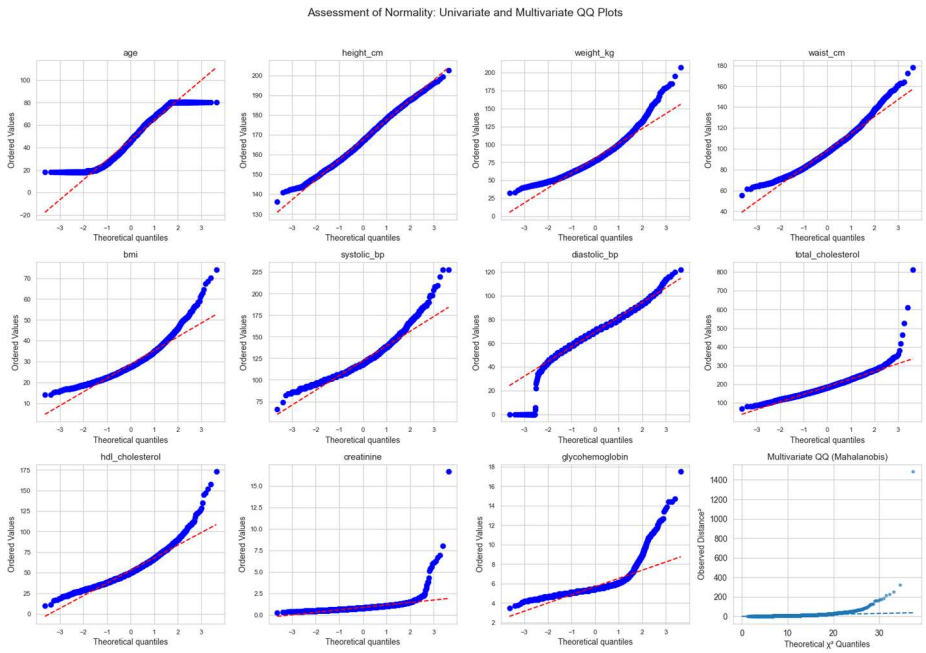


Figure 3. Univariate and multivariate QQ plots for assessing normality.
Most variables show clear deviations from the reference line, indicating right-skewed or heavy-tailed distributions. The multivariate QQ plot also displays a heavy tail, consistent with the presence of individuals with jointly elevated health indicators. These patterns are confirmed by Mardia's test, which strongly rejects multivariate normality. The violation of normality assumptions motivates the use of ICA (which requires non-Gaussianity) and suggests caution when applying covariance-based methods such as PCA, FA, and GMM, as their results may be influenced by outliers or tail behavior.

Since most multivariate methods either need MVN or non normality assumptions, we first test the normality assumption. The QQ plots above (Figure 3) show deviations from the diagonal lines, indicates that most features do not follow a univariate Gaussian distribution. Particularly, weight, waist, bmi, and the multivariate normal qq plot are convex, the corresponding features are right skewed, which indicates the presence of individuals with very high body measurements, aligning with the outlier patterns observed in the boxplots (Figure 1).

The heavy multivariate tail suggests the presence of latent subpopulations with jointly elevated health indicators, which motivates the use of clustering methods such as GMM. In addition, by conducting Mardia's test, we obtain Mardia's Skewness (b1p) = 340.44603644121526 with p-value = 0.0, and Mardia's Kurtosis (b2p) = 802.530076113545 with p-value = 0.0. Since both p-values are less than 0.05, we reject the null hypothesis that the dataset follows a multivariate normal distribution. Thus, we conclude that the dataset does not satisfy the assumption of multivariate normality. This result isn't surprising since there are many people with extreme health indicators and represented as outliers in the dataset, which would drive the group distribution away from multivariate normal distribution. The assumption test is important because methods such as Factor Analysis and Gaussian Mixture Model often rely to varying degrees on multivariate normality assumptions, while Independent Component Analysis (ICA) needs non-Gaussian. In particular, ICA assumes that the underlying sources are non-Gaussian and statistically independent, with at most one component being Gaussian. Therefore, the observed departure from multivariate normality provides justification for applying ICA to this dataset. In addition, the violation of MVN may distort PCA loadings, which suggests conducting robust PCA with MVN.

Model Specification and Interpretation:

Based on the observations above, several features are highly correlated, indicating redundancy among variables. Therefore, Principal Component Analysis (PCA) is applied to reduce the dimensionality of the dataset while retaining most of the variability. PCA achieves this by constructing orthogonal principal components that capture maximal variance.

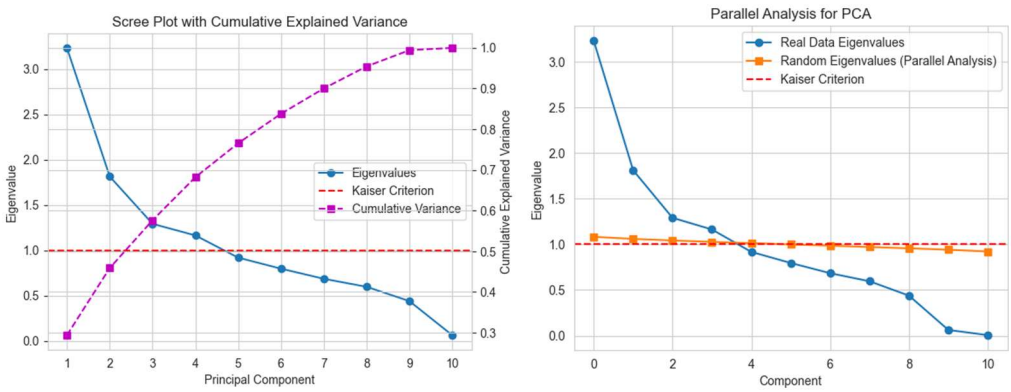


Figure 4. PCA scree plot and parallel analysis.
The scree plot (left) shows that the first four components have eigenvalues above 1 and together explain about 70% of the variance. Parallel analysis (right) confirms that only these four components exceed the eigenvalues expected from random data. Both criteria support retaining four principal components to summarize the correlated health indicators while reducing redundancy for later analyses.

Principal Component Analysis (PCA)

Based on the strong correlations observed among subsets of variables—particularly the body measurement block (weight, waist, BMI)—Principal Component Analysis (PCA) was first applied to reduce dimensionality and remove redundancy. PCA constructs orthogonal components that capture maximal variance, making it good for summarizing correlated features into a smaller set of variables.

To determine the number of components, we plotted the scree plot (Figure 4 left) which doesn't show a clear elbow. According to the Kaiser criterion, the first 4 principal components have eigenvalues greater than 1. We also calculated eigenvalue ratios for the first 7 gaps and obtained the result 1.7832, 1.4043, 1.11, 1.2673, 1.1529, 1.1632, 1.1486. Since the eigenvalue ratio are considered stable when larger than 2, apparently this result is not very informative. and cumulative variance for the first 4 PCs is around 70%. To be more accurate, we conducted parallel analysis (Figure 4 right) which suggested us to keep 4 pcs.

Considering multiple criteria, it is reasonable to keep 4 principal components. Thus, the original 11 dimensions dataset was reduced to 4 dimensions while still preserving 70% the variability in the data. This lower-dimensional representation provides a more stable and space for subsequent clustering analysis, as it removes redundant information that could otherwise distort distance-based or model-based methods such as GMM.

Factor Analysis (FA)

To further investigate whether the observed correlations can be explained by latent factors, Factor Analysis (FA) was applied. The suitability of FA was assessed using the Kaiser–Meyer–Olkin (KMO) statistic and Bartlett's test. The KMO value of 0.53 falls in the "miserable" range, suggesting that the overall correlation structure is weak and that the data may not be well-explained by a small number of common factors. This is consistent with earlier observations that only a subset of variables (e.g., body measurements) exhibit strong correlations, while many others are relatively independent.

However, Bartlett's test is highly significant ($p \approx 0$), indicating that the correlation matrix is not an identity matrix and that some shared structure exists. This suggests that while FA may not fully capture the data with a small number of factors, it can still provide partial insights into latent relationships.

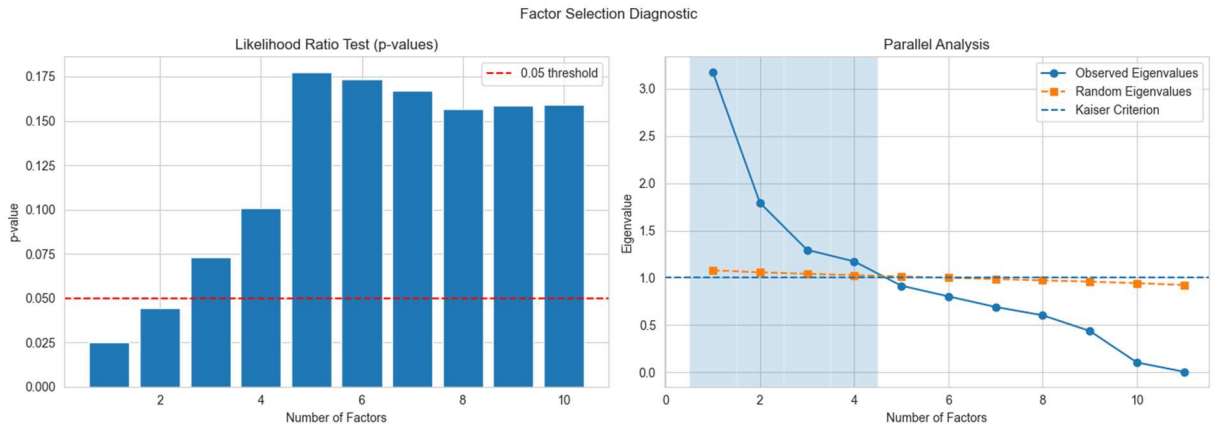


Figure 5. Factor selection diagnostics using likelihood ratio testing and parallel analysis.
The likelihood ratio test (left) shows that models with fewer than four factors are rejected at the 0.05 level, while parallel analysis (right) indicates that only the first few observed eigenvalues exceed those expected from random data. Together, these diagnostics suggest retaining around four to five factors as the most reasonable balance between statistical support and interpretability.

The number of factors was determined using parallel analysis (Figure 5 right) and likelihood ratio testing (Figure 5 left). Both methods suggest retaining at least four factors. However, inspection of the factor loading patterns reveals that a five-factor solution provides an additional interpretable dimension dominated by HDL cholesterol. Therefore, five factors were retained to balance statistical criteria with interpretability.

From the Factor Loadings Heatmap (Figure 6 left)

Factor 1 (Body Mass / Obesity): strong loadings on weight, waist, and BMI

Factor 2 (Aging / Metabolic Risk): dominated by age, with moderate contributions from systolic blood pressure and glycohemoglobin

Factor 3 (Body Size): primarily height-related

Factor 4 (Blood Pressure): driven by systolic and diastolic blood pressure

Factor 5 (HDL cholesterol): dominated by HDL cholesterol

Despite these interpretable dimensions, the weak KMO suggests that FA captures only part of the underlying structure, and its results should be interpreted cautiously.

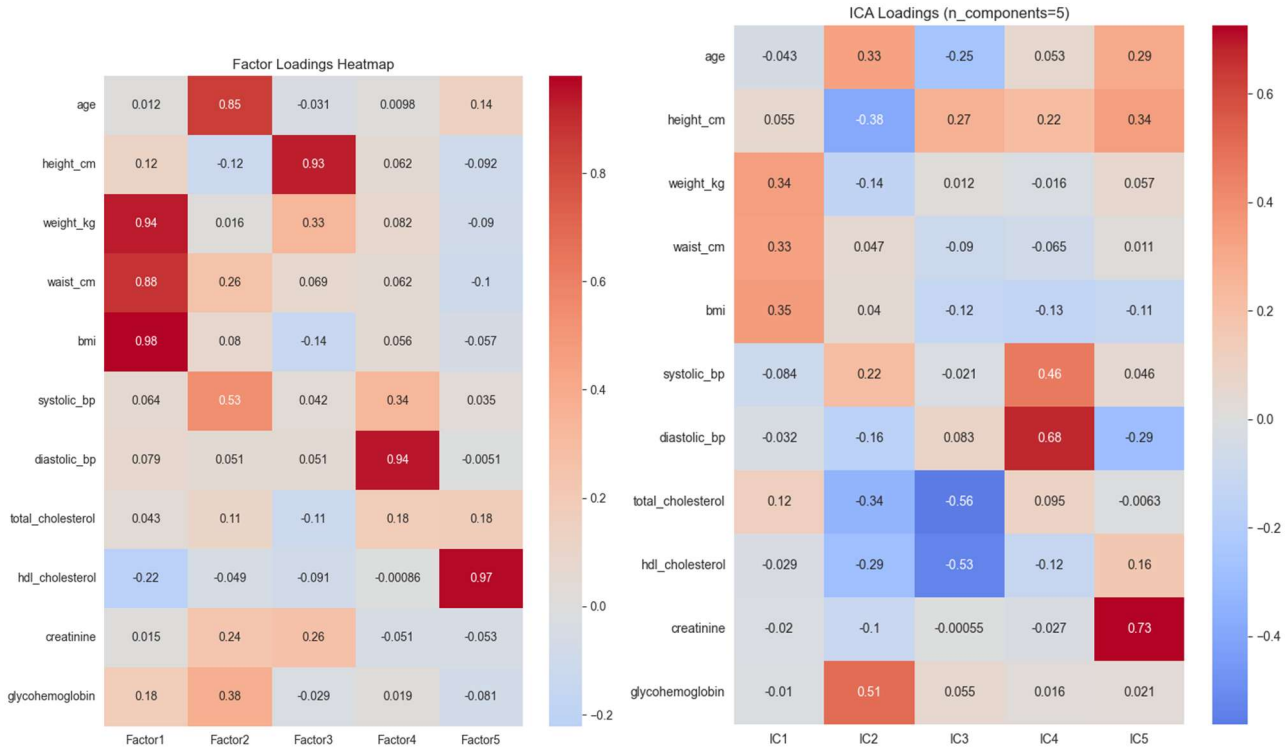


Figure 6. Comparison of factor loadings and ICA loadings.
The left heatmap shows the five-factor solution from FA, revealing interpretable dimensions related to body mass, aging/metabolic risk, body size, blood pressure, and HDL cholesterol. The right heatmap displays the five independent components from ICA, which identify obesity, metabolic, cholesterol, blood-pressure, and physiological function axes. While FA and ICA share similarities in the body-mass and blood-pressure dimensions, the remaining components differ, reflecting that FA captures shared covariance structure whereas ICA extracts statistically independent sources.

Independent Component Analysis (ICA)

Given the strong evidence of non-Gaussianity, ICA was applied to identify statistically independent sources of variation. Unlike PCA and FA, which rely on covariance structure, ICA exploits higher-order moments and is particularly suited for non-Gaussian data.

We conducted Mardia's Test to check the suitability to apply ICA by calculating kurtosis for keeping different numbers of ICA components and get the result [0.5021, 16.9781,15.3125, 30.0652, 27.8296, 40.395, 68.0711]. The data is non-Gaussian after IC2. Since Kurtosis is sensitive to outliers, where there are many outliers in this health dataset as we analysed in the beginning of the report, we also calculated the negentropy to consolidate our belief and got the result [0.0, 0.0012, 0.0013, 0.0024, 0.0029, 0.0052, 0.0084], the first 3 independent components are close to Gaussian and after IC4 its near non gaussian. Also to be consistent with the Factor analysis result, we keep 5 independent components.

The ICA components are shown in Figure 6 right:

IC1 (Obesity axis): weight, waist, BMI

IC2 (Metabolic axis): glycohemoglobin and age, with negative cholesterol loading

IC3 (Cholesterol axis): total and HDL cholesterol

IC4 (Blood pressure axis): systolic and diastolic BP

IC5 (physiological function): creatinine, age, height

These components suggest that the data is generated by multiple statistically independent sources rather than a small number of correlated latent factors.

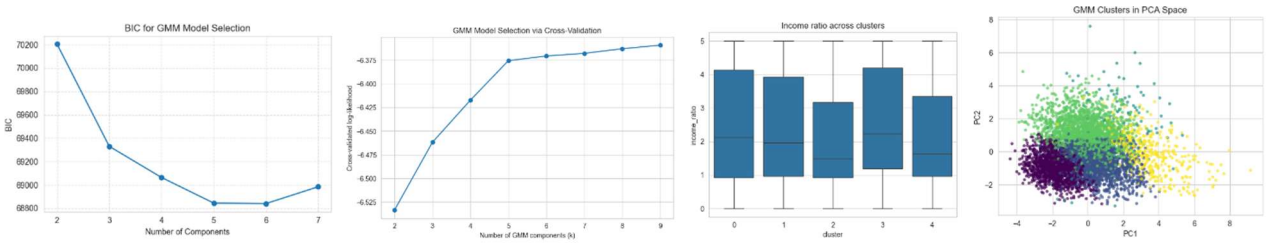
Comparison of FA and ICA (Cross-Method Synthesis)

From the correlation calculation of FA and ICA loadings, FA1 and IC1 has the largest correlation 0.98, they both represent the similar body measurements. FA4 and IC4 have the second highest correlation 0.86, both represent the blood pressure. These results indicating that two methods reveal similar overall underlying structures. For other components, there are no clear relationships which real the differences between FA and ICA.

However, other components show weak correspondence, reflecting fundamental differences between the methods. FA assumes shared covariance structure, while ICA seeks statistical independence.

ICA stability analysis shows relatively low stability ([0.33 0.304 0.294 0.319 0.31]), indicating that the independent components are not very stable. This further supports the interpretation that the underlying structure is complex and not fully captured by a single model.

Gaussian Mixture Model (GMM)



cluster	age	height_cm	weight_kg	waist_cm	bmi	systolic_bp	diastolic_bp	total_cholesterol	hdl_cholesterol	creatinine	glycohemoglobin
0	29.35	167.77	65.42	82.33	23.16	109.40	64.16	169.54	56.17	0.82	5.16
1	38.97	170.39	90.51	103.97	31.22	117.75	72.33	193.06	47.25	0.86	5.40
2	63.47	165.19	80.75	103.39	29.41	146.81	53.48	191.92	48.59	2.25	7.89
3	63.82	164.43	74.15	97.12	27.35	132.78	71.27	195.58	56.61	0.93	5.94
4	53.65	167.45	119.82	129.25	42.58	128.33	71.07	185.48	45.71	0.89	6.89

education cluster	1.0	2.0	3.0	4.0	5.0	7.0	9.0
0	0.03	0.11	0.19	0.32	0.35	0.0	0.0
1	0.05	0.15	0.23	0.33	0.24	0.0	0.0
2	0.21	0.22	0.21	0.22	0.14	0.0	0.0
3	0.11	0.14	0.23	0.28	0.24	0.0	0.0
4	0.12	0.11	0.26	0.36	0.15	0.0	0.0

gender cluster	Female	Male
0	0.57	0.43
1	0.43	0.57
2	0.43	0.57
3	0.54	0.46
4	0.59	0.41

Figure 7. Gaussian Mixture Model selection and cluster characterization.
The top row shows model selection results: BIC identifies 5–6 components as optimal, and cross-validated log-likelihood shows a clear improvement up to five clusters. The PCA-space visualization confirms that the five-cluster solution forms well-separated groups. The bottom row summarizes cluster characteristics, including health indicators, income, education, and gender distributions. The clusters correspond to meaningful subpopulations—young healthy adults, middle-aged high-risk individuals, two distinct older groups, and an obese middle-aged group—indicating that the latent structures identified earlier align with real demographic and health differences.

To investigate whether these latent structures correspond to distinct subpopulations, Gaussian Mixture Models (GMM) were applied for clustering. Since the original variables exhibit strong correlations and non-Gaussianity, clustering was performed in the PCA-reduced space, where redundancy and noise are reduced, improving model stability and interpretability.

Although the overall data distribution is non-Gaussian, GMM assumes only approximate Gaussianity within each cluster. In the context of health data, this is a reasonable approximation, as each cluster may represent a subgroup with relatively homogeneous characteristics.

The optimal number of clusters was selected using the Bayesian Information Criterion (BIC). From the number of clusters vs BIC plot (Figure 7), the best cluster number is 5 or 6. Given the marginal improvement from 5 to 6 clusters, a 5-cluster solution was chosen for interpretability. Cross-validation further supports this choice, showing a clear improvement from 4 to 5 clusters. We also conducted cross validation, and it shows that from 4 clusters to 5 clusters there is a clear improvement on the likelihood.

From the GMM result (Figure 7), we can see

Cluster 0 is younger people in their 20~30s, they have relatively lower average body measurements values, lower cholesterol, and healthier blood pressure range. Compared to other clusters, this group of people are generally healthier.

Cluster 1 is for people around 40s with higher body weight, bmi, blood pressure and cholesterols.

Cluster 2 and Cluster 3 is the group of older people in their 60s. Cluster 2 are elder people but relatively unhealthy. They have the most extreme blood pressure-higher systolic bp and lower diastolic bp on average, and highest creatine and glycohemoglobin in the five group. Cluster 3 are older people but healthier.

Cluster 4 are people in their 50s and tend to have obesity. They have the highest weight, waist circumference, and BMI.

We plotted 5 clusters on PC1 vs PC2 2-dimensional space (Figure 7). We can clearly see there are 5 clusters, with some separation since we conducted GMM on 4-dimensional PCA space.

We also plotted the boxplot of income based on different clusters and made a table of education distribution based on different clusters. It is noticeable that within group of elder people (Cluster 2 and Cluster 3), Cluster 3 has relatively higher income and higher education level than Cluster 2, and they are healthier. Thus there may be some connection between society status and healthy.

We can also see from the table of gender vs clusters, there is no clear separation of genders in different classes.

Robust Estimation:

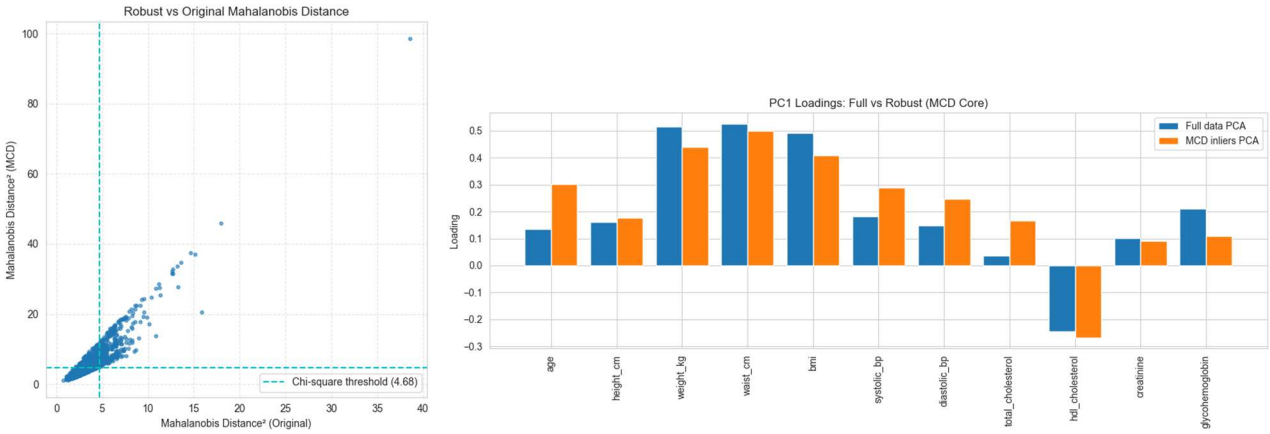


Figure 8. Robust Mahalanobis distances and comparison of PCA loadings.
The left plot compares classical and MCD-based Mahalanobis distances, showing that many outliers are masked under the classical covariance but revealed by the robust estimator. The right plot compares PCA loadings from the full dataset with those from the MCD inliers; the dominant loading pattern remains stable across both versions. These results indicate that while outliers strongly distort distance measures, the main latent structure captured by PCA is robust to extreme observations.

Given the number of outliers we have, robust estimation was conducted to assess whether the main findings are sensitive to extreme observations. Since the ratio $n/p \approx 300$, the sample covariance matrix is well-conditioned, and shrinkage methods are not necessary. Instead, the Minimum Covariance Determinant (MCD) estimator was applied to obtain a robust estimate of location and covariance.

We conducted MCD and checked outliers again based on the chi square 0.975 threshold and get 1354 outliers, substantially more than those detected using the classical covariance estimator. From the DD plot (Figure 8), we can see that a small portion of outliers are swamped and a large portion of outliers are masked, where extreme observations distort the covariance structure and prevent other outliers from being detected.

This result has two important implications. First, it provides further evidence that the dataset deviates strongly from multivariate normality. Second, we want to know whether the main structures identified earlier are genuine or driven by these extreme observations.

Impact on PCA and Latent Structure

Robust PCA based on MCD covariance yields component loadings that are consistent with those obtained from the full dataset. In particular, the first principal component continues to load strongly on BMI, waist circumference, weight, and cardiovascular indicators, representing a dominant "metabolic risk" axis.

The stability of this component suggests that the primary direction of variation is not an artifact of extreme values but rather reflects a genuine population-level structure. In other words, while outliers affect the distributional shape, they do not fundamentally alter the main variance structure captured by PCA.

Impact on Factor Analysis

Applying FA to the MCD-based inliers produces factor loadings that are broadly similar to those obtained from the full dataset. However, this stability does not imply that FA is well-suited for the data. Rather, it suggests that the limitations of FA arise primarily from weak correlation structure and violation of model assumptions (e.g., multivariate normality and common factor structure), rather than from the presence of outliers.

This reinforces the earlier conclusion that FA provides only a partial and potentially fragile representation of the underlying structure.

Impact on Clustering (GMM)

Clustering results obtained from GMM in the PCA space remain largely unchanged when using robust covariance estimates or restricting to MCD inliers. The same cluster profiles—young healthy individuals, middle-aged high-risk groups, and elderly subgroups—are consistently recovered.

This robustness indicates that the identified clusters reflect genuine subpopulation structure rather than being driven by a small number of extreme observations. Instead, the clusters appear to capture systematic variation across the population, consistent with the heavy-tailed and heterogeneous nature of the data.

Across all robustness checks, PCA and GMM remained stable when using MCD-based inliers or robust covariance, indicating that the main latent structure and cluster solution are not because of outliers. FA remained unstable even under robust estimation, suggesting that their limitations arise from model assumptions rather than outliers.

Discussion:

This analysis aimed to identify the underlying structure of health indicators and their relationship with demographic factors in the NHANES dataset. Across multiple methods, we know that the data is not well described by a small number of global latent factors but instead reflects a combination of a dominant variance axis and heterogeneous subpopulations.

PCA reveals a stable primary dimension corresponding to overall metabolic risk, driven by body measurements and cardiovascular indicators. This structure is consistently recovered across robust estimation, indicating that it is a genuine population-level pattern rather than an artifact of outliers. In contrast, Factor Analysis provides only a partial representation due to weak global correlations and violation of its modeling assumptions.

ICA and GMM further refine this picture. ICA suggests that the data is generated by multiple statistically independent physiological processes, while GMM identifies distinct population subgroups that differ in age, metabolic risk, and health status. Linking these clusters to demographic variables shows that healthier profiles, particularly among older individuals, are associated with higher socioeconomic status, while gender plays a limited role.

Taken together, these results suggest that population health is best understood not through a single latent structure, but through the interaction of independent biological processes and heterogeneous subpopulations. This explains both the strong non-normality of the data and the limitations of methods that rely on global assumptions such as multivariate normality.

AI Attribution and Acknowledgements

Generative AI (ChatGPT) was used for proofreading, debugging code, and improving the clarity of the text. The author independently conducted all analyses, made all modeling decisions, and verified all results.